UNIT 1

DC & AC Circuits

Topics: Circuit elements (R, L, C) • Ohm's law • KCL & KVL • Series/parallel circuits • Superposition theorem • AC fundamentals • RMS, average, form & peak factor • Phasors in R, L, C • Impedance • Active, reactive & apparent power • Power factor

START HERE
Big picture: Electricity is the flow of charge. Every electrical circuit has a **source** (battery/supply), a **path** (wires) and a **load** (bulb, motor, heater). Unit I teaches the rules that tell us how much voltage and current each part gets.

1 DC CIRCUITS

1.1 Circuit Elements: R, L and C

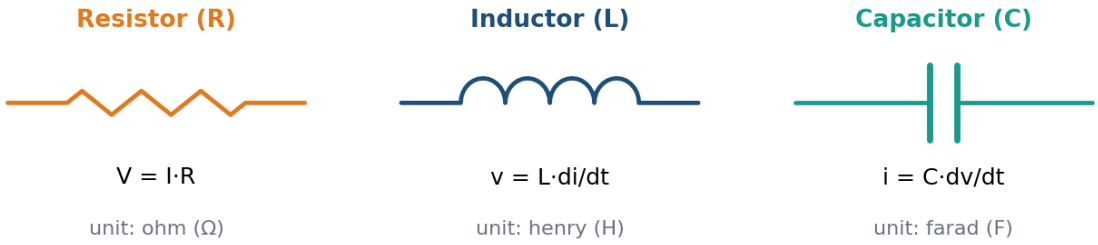


Fig 1.1 Symbols of the three passive elements

Element	What it does	Energy	Behaviour in steady DC
R Resistor	Opposes current; turns electrical energy into heat	Dissipated: $P = I^2R$	Acts as resistance R
L Inductor	Opposes *change* of current (magnetic field)	Stored: $\frac{1}{2}LI^2$ (joule)	Short circuit (plain wire)
C Capacitor	Opposes *change* of voltage (electric field)	Stored: $\frac{1}{2}CV^2$ (joule)	Open circuit (blocks DC)

DEFINITION
Resistance of a wire: $R = \rho l/A$, where ρ = resistivity (Ω·m), l = length, A = cross-section area. Longer/thinner wire → more resistance.

APPLICATIONS
R: heaters, bulbs, fuses, voltage dividers. **L:** chokes, transformers, motors, relays. **C:** filters, power-supply smoothing, fan/motor starting, tuning circuits.

1.2 Ohm’s Law and its Limitations

DEFINITION

Ohm’s Law: At constant temperature, the current through a conductor is directly proportional to the voltage across it.

$$V = I \times R \Rightarrow I = V / R \Rightarrow R = V / I$$

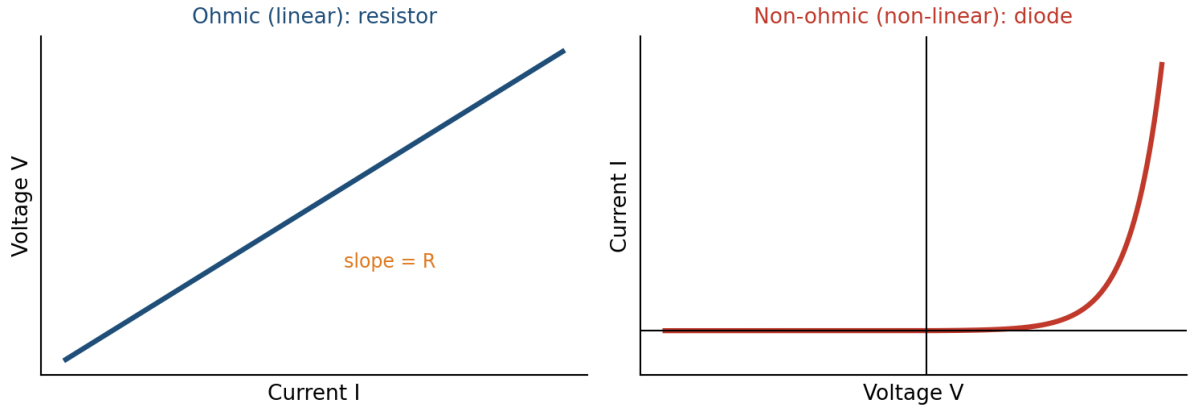


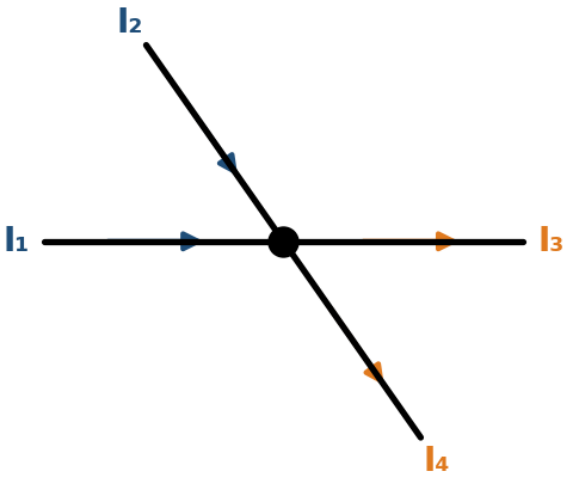
Fig 1.2 Ohmic (straight line) vs non-ohmic (curved) behaviour

Limitations of Ohm’s law

- Not valid if the **temperature** changes (resistance of metals rises with heat).
- Not applicable to **non-linear devices**: diodes, transistors, thermistors, SCRs.
- Not applicable to electrolytes, arc lamps, vacuum tubes and other **unilateral** elements.

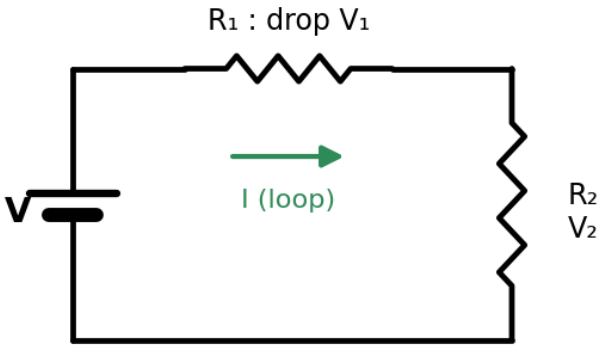
1.3 Kirchhoff’s Laws

Kirchhoff’s Current Law



KCL: $I_1 + I_2 = I_3 + I_4$
(current in = current out)

Kirchhoff’s Voltage Law



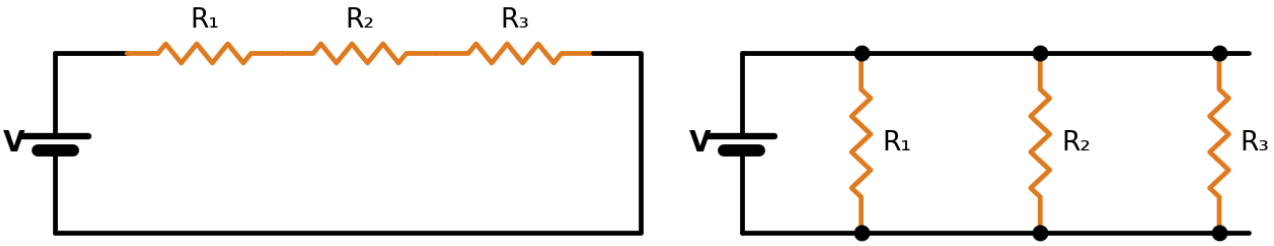
KVL: $V - V_1 - V_2 = 0$
(sum of voltages around a loop = 0)

Fig 1.3 KCL at a junction (left) and KVL around a loop (right)

	KCL (Current Law)	KVL (Voltage Law)
Statement	Algebraic sum of currents at a node = 0 (current in = current out)	Algebraic sum of voltages (rises and drops) in a closed loop = 0
Based on	Conservation of charge	Conservation of energy
Equation	$\Sigma I_{in} = \Sigma I_{out}$	$\Sigma V = 0$ (Source V = sum of drops)

EXAM TIP
Steps for KVL: (1) mark current direction in each branch, (2) choose a loop direction, (3) write drop = +IR if you go *along* the current, (4) a source is + when you pass from − to +, (5) set the sum = 0 and solve.

1.4 Series, Parallel and Series-Parallel Circuits



SERIES: same current I

R_eq = R₁ + R₂ + R₃

PARALLEL: same voltage V

1/R_eq = 1/R₁ + 1/R₂ + 1/R₃

Fig 1.4 Series and parallel connections of resistors

Property	Series	Parallel
Current	Same in all (I)	Divides: I = I₁ + I₂ + I₃
Voltage	Divides: V = V₁ + V₂ + V₃	Same across all (V)
Equivalent R	R₁ + R₂ + R₃ (larger than the biggest)	1/R₁ + 1/R₂ + 1/R₃ (smaller than the smallest)
Two resistors	—	R_{eq} = R₁R₂ / (R₁ + R₂)
Use	Fuse, switch, string lights	Household wiring (each device gets full V)

SOLVED EXAMPLE
Q1. 12 Ω, 6 Ω and 4 Ω are in parallel across 12 V. Find R_{eq} and branch currents.
$1/R_{eq}$ = 1/12 + 1/6 + 1/4 = 6/12 = 0.5 ⇒ **R_{eq} = 2 Ω**; total I = 12/2 = **6 A**. Branches: 12/12 = **1 A**, 12/6 = **2 A**, 12/4 = **3 A** (sum = 6 A ✓).
Q2 (series-parallel). 4 Ω in series with (6 Ω ∥ 12 Ω) across 24 V. 6∥12 = 4 Ω ⇒ R_{total} = 8 Ω ⇒ I = **3 A**. Voltage across the parallel part = 3 × 4 = 12 V ⇒ currents 2 A and 1 A.

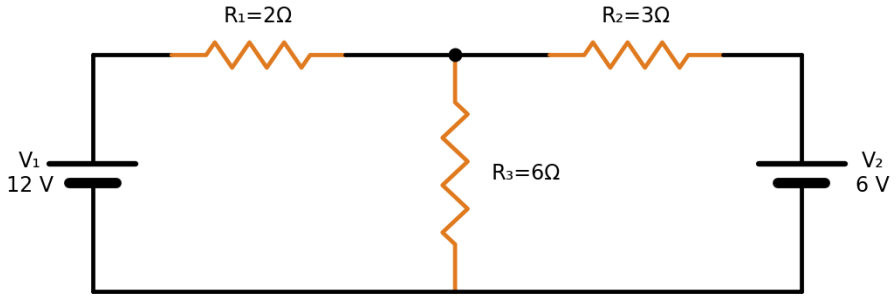
HANDY SHORTCUTS
Voltage divider (series): V₁ = V × R₁/(R₁+R₂) | **Current divider** (two parallel branches): I₁ = I × R₂/(R₁+R₂)

REMEMBER
In a **series** circuit the *largest* resistor gets the *largest* voltage; in a **parallel** circuit the *smallest* resistor carries the *largest* current. Household appliances are connected in **parallel** so that each gets full supply voltage and works independently.

1.5 Superposition Theorem

DEFINITION

In a **linear** circuit with more than one source, the current in any branch equals the **algebraic sum** of the currents produced by each source acting **alone**, with the other sources replaced by their internal resistance (**voltage source → short circuit; current source → open circuit**).



Find the current through R₃

Fig 1.5 Example circuit for superposition

SOLVED EXAMPLE

Step 1 (V_1 = 12 V only, V_2 shorted): $R_2 \| R_3 = 3\times6/9 = 2\ \Omega$; total = 2 + 2 = 4 Ω; I = 3 A; voltage across $R_3 = 3 \times 2 = 6$ V ⇒ $I_3' = 6/6 =$ **1 A**.

Step 2 (V_2 = 6 V only, V_1 shorted): $R_1 \| R_3 = 2\times6/8 = 1.5\ \Omega$; total = 3 + 1.5 = 4.5 Ω; I = 1.333 A; voltage across $R_3 = 1.333 \times 1.5 = 2$ V ⇒ $I_3'' = 2/6 =$ **0.333 A**.

Step 3 (add): $I_3 = 1 + 0.333 =$ **1.33 A** (downward). Check by node analysis: $V_A = 8$ V, 8/6 = 1.33 A ✓.

REMEMBER

Superposition works only for **linear** circuits and for current/voltage — **not for power** (power ∝ I^2, which is not linear).

2 AC CIRCUITS

2.1 AC Fundamentals

Alternating current (AC) changes its magnitude and direction periodically; the usual shape is a **sine wave**. Mains supply in India: **230 V, 50 Hz**. AC is preferred because voltage can be easily stepped up/down by transformers for efficient transmission.

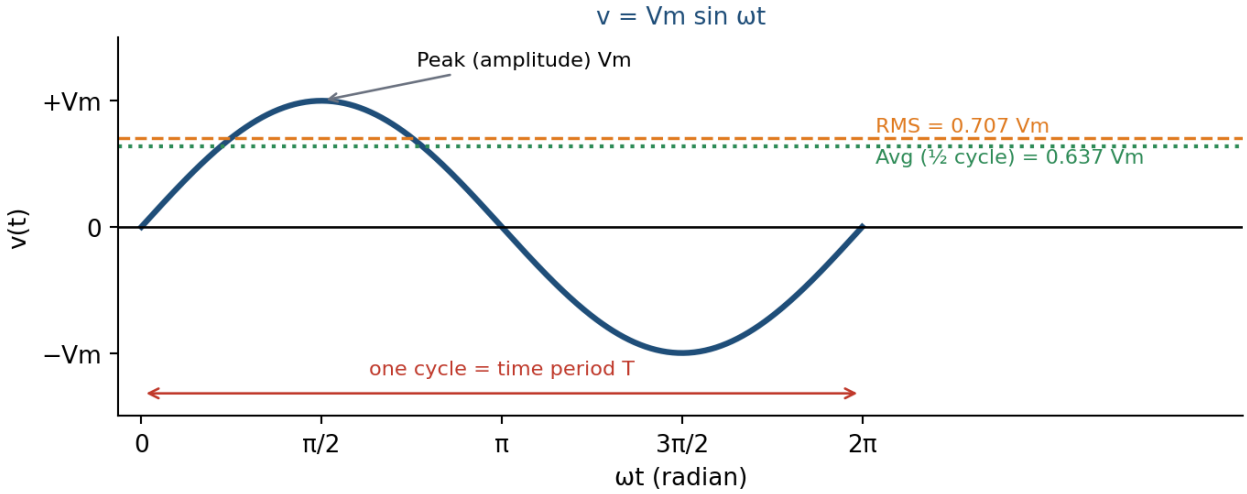


Fig 2.1 Sinusoidal waveform and its important values

$v = V_m \sin(\omega t)$ $i = I_m \sin(\omega t \pm \varphi)$ $\omega = 2\pi f = 2\pi/T$

Term	Meaning	Unit
Amplitude (peak)	Maximum value V_m or I_m	V or A
Time period T	Time for one complete cycle	s
Frequency f	Cycles per second, f = 1/T	Hz
Phase	Angular position of the wave at a given instant (ωt ± φ)	rad / degree
Phase difference φ	Angle by which one wave leads/lags another (Fig 2.2)	degree

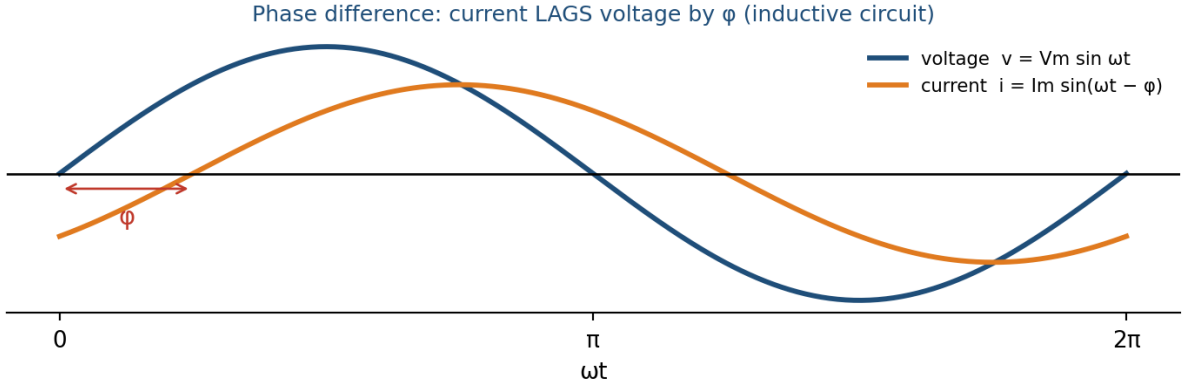


Fig 2.2 Phase difference between voltage and current

2.2 Average, RMS, Form Factor and Peak Factor

Quantity	Definition	Sine wave
Average value	Mean of instantaneous values over a *half* cycle (full-cycle average = 0)	$V_{av} = 2V_m/\pi =$ **0.637 V_m**
RMS (effective) value	Square root of mean of squares; the DC value giving the same heating effect	$V_{rms} = V_m/\sqrt{2} =$ **0.707 V_m**
Form factor	RMS value / Average value	**1.11**
Peak factor (crest)	Peak value / RMS value	**1.414**

SOLVED EXAMPLE

Q3. v = 311 sin 314t. Find V_{rms}, f and T.
$V_m = 311$ V ⇒ $V_{rms} = 0.707 \times 311 =$ **220 V**; ω = 314 ⇒ f = 314/2π = **50 Hz**; T = 1/50 = **20 ms**. Average (half-cycle) = 0.637 × 311 = 198 V.

2.3 AC through R, L and C (Phasor Diagrams)

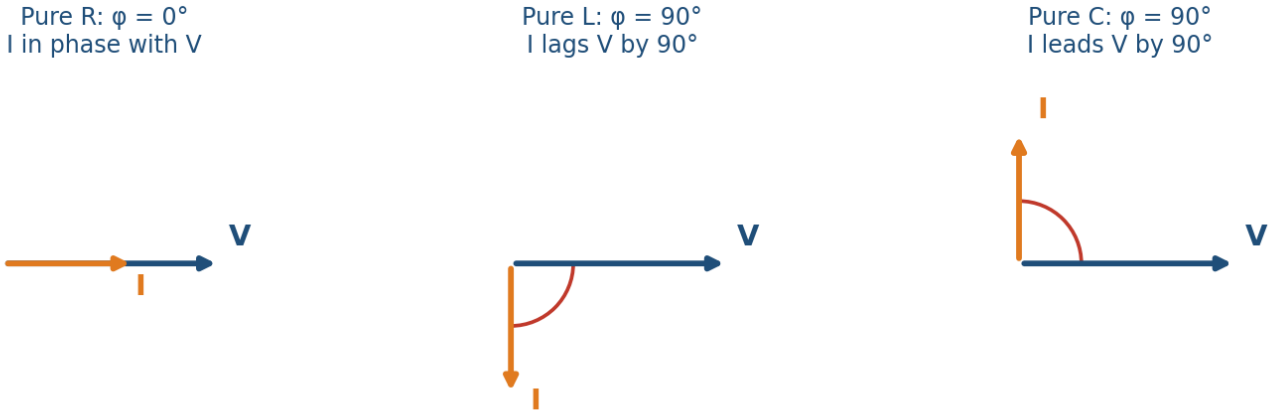


Fig 2.3 Phasor diagrams (voltage V is the reference along the x-axis)

Circuit	Opposition	Phase relation	Memory trick
Pure R	Z = R	I in phase with V (φ = 0°)	—
Pure L	$X_L = \omega L = 2\pi fL$	I **lags** V by 90°	"ELI": E before I in L
Pure C	$X_C = 1/\omega C = 1/2\pi fC$	I **leads** V by 90°	"ICE": I before E in C

Concept of Impedance (Z): the total opposition of a circuit to AC, combining resistance and reactance. For a series R-L-C circuit:

$$\mathbf{Z = \sqrt{[\ R^2 + (X_L - X_C)^2\ ]}\ \tan \varphi = (X_L - X_C) / R\ I = V / Z}$$

If $X_L > X_C$ the circuit is **inductive** (current lags); if $X_C > X_L$ it is **capacitive** (current leads); if $X_L = X_C$ it is **resonance** (Z = R, minimum).

SOLVED EXAMPLE

Q4. Find X_L of a 0.1 H coil and X_C of a 100 μF capacitor at 50 Hz.
$X_L = 2\pi \times 50 \times 0.1 =$ **31.4 Ω**; $X_C = 1/(2\pi \times 50 \times 100\times10^{-6}) =$ **31.8 Ω**. Note: X_L *increases* with frequency, X_C *decreases* with frequency (a capacitor passes high frequencies, an inductor blocks them).

APPLICATIONS

Series RLC resonance occurs at $f_0 = 1/(2\pi\sqrt{LC})$, where $X_L = X_C$, Z = R (minimum), current is maximum and pf = 1 — used in radio tuning.

2.4 Active, Reactive and Apparent Power; Power Factor

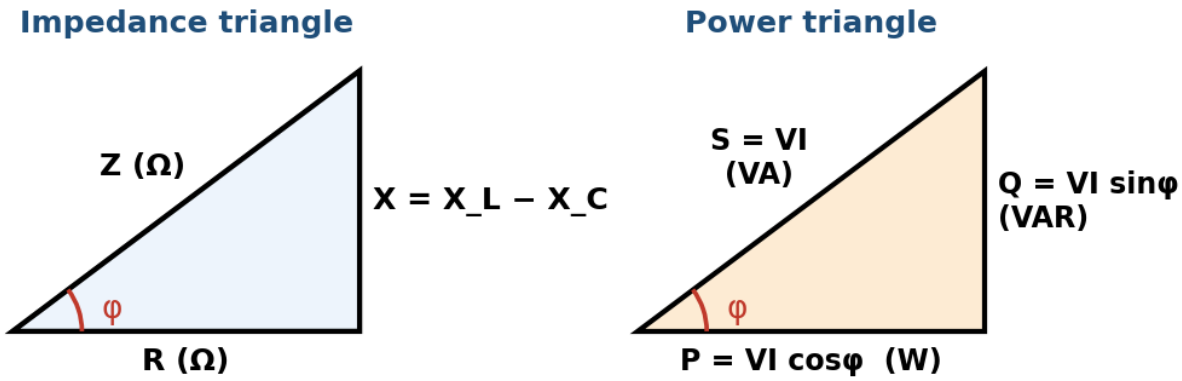


Fig 2.4 Impedance triangle and power triangle

Power	Symbol & formula	Unit	Meaning
Active (true)	P = VI cosφ = I²R	watt (W)	Real work done (heat, light, motion)
Reactive	Q = VI sinφ = I²X	VAR	Energy swapped between source and L/C; no work
Apparent	S = VI = I²Z	VA	Total power supplied; S² = P² + Q²

DEFINITION

Power factor = cos φ = R/Z = P/S (lies between 0 and 1; lagging for inductive loads, leading for capacitive loads). A **low power factor** means higher current for the same power → more losses and larger cables; it is improved by connecting **capacitors** across inductive loads.

SOLVED EXAMPLE

Q5. A series circuit has R = 30 Ω and X_L = 40 Ω across 230 V, 50 Hz. Find Z, I, pf, P, Q, S.
Z = √(30² + 40²) = **50 Ω**; I = 230/50 = **4.6 A**; φ = $\tan^{-1}(40/30)$ = 53.1°; pf = 30/50 = **0.6 lagging**.
P = 230 × 4.6 × 0.6 = **635 W** (check: I²R = 21.16 × 30 = 635 ✓); Q = 230 × 4.6 × 0.8 = **846 VAR**; S = 230 × 4.6 = **1058 VA**.

3 QUICK REVISION & EXAM CORNER

Topic	Key formulas
Ohm’s law	$V = IR$
Series / Parallel	$R_{eq} = \Sigma R$ / $1/R_{eq} = \Sigma(1/R)$
KCL / KVL	$\Sigma I = 0$ at node / $\Sigma V = 0$ in loop
AC wave	$\omega = 2\pi f$, $T = 1/f$, $V_{rms} = 0.707V_m$, $V_{av} = 0.637V_m$, FF = 1.11, PF(peak) = 1.414
Reactance & Z	$X_L = 2\pi fL$, $X_C = 1/2\pi fC$, $Z = \sqrt{(R^2 + X^2)}$
Power	$P = VI\cos\varphi$, $Q = VI\sin\varphi$, $S = VI$, $pf = \cos\varphi$

Short questions (2 marks)

- State Ohm’s law. Mention two limitations.
- State KCL and KVL.
- Define RMS value, form factor and peak factor of a sine wave.
- What is the phase relation between V and I in a pure inductor and a pure capacitor?
- Define impedance and power factor.
- Why is the RMS value used to rate AC supplies?

Long questions (5–10 marks)

- Explain superposition theorem with a neat example circuit.
- Derive RMS and average values of a sinusoidal current.
- Draw phasor diagrams for pure R, L and C circuits and explain.
- Explain active, reactive and apparent power with the power triangle.
- Three resistors 5 Ω, 10 Ω and 20 Ω are connected in (a) series and (b) parallel across 40 V. Find total current in each case. *(Ans: 1.14 A; 14 A)*

Practice numericals (with answers)

No.	Problem	Answer
1	Two resistors 6 Ω and 3 Ω are in parallel across 18 V. Find total current.	9 A (R_{eq} = 2 Ω)
2	Find RMS and average value of a sine wave with peak 200 V.	141.4 V; 127.4 V
3	R = 8 Ω in series with X_L = 6 Ω across 100 V. Find I and pf.	10 A; 0.8 lag
4	A load draws 5 A at 230 V with pf 0.8. Find P and Q.	920 W; 690 VAR
5	Find the series equivalent of 10 Ω, 15 Ω and 25 Ω, and current at 100 V.	50 Ω; 2 A

Quick MCQs

Question	Options	Ans
1. Unit of inductance is	(a) farad (b) henry (c) ohm (d) weber	b
2. Which law is based on conservation of charge?	(a) KVL (b) Ohm (c) KCL (d) Faraday	c
3. Form factor of a sine wave is	(a) 1.414 (b) 1.11 (c) 0.707 (d) 0.637	b
4. In a pure capacitor, current ... voltage by 90°	(a) lags (b) leads (c) is in phase with (d) is zero	b
5. Power factor of a pure inductor is	(a) 1 (b) 0.5 (c) 0 (d) infinity	c
6. Voltage source is replaced by ... in superposition	(a) open circuit (b) short circuit (c) resistor (d) capacitor	b

UNIT 2

Machines & Measuring Instruments

Topics: DC generator • DC motor • Single-phase transformer • Three-phase induction motor • Alternator • Applications of machines • PMMC instrument • Moving-iron instrument • Wheatstone bridge

START HERE

Big picture: An **electrical machine** converts energy. A **generator** turns mechanical energy into electrical; a **motor** does the reverse; a **transformer** changes voltage level with no moving parts. **Measuring instruments** let us read current, voltage and resistance.

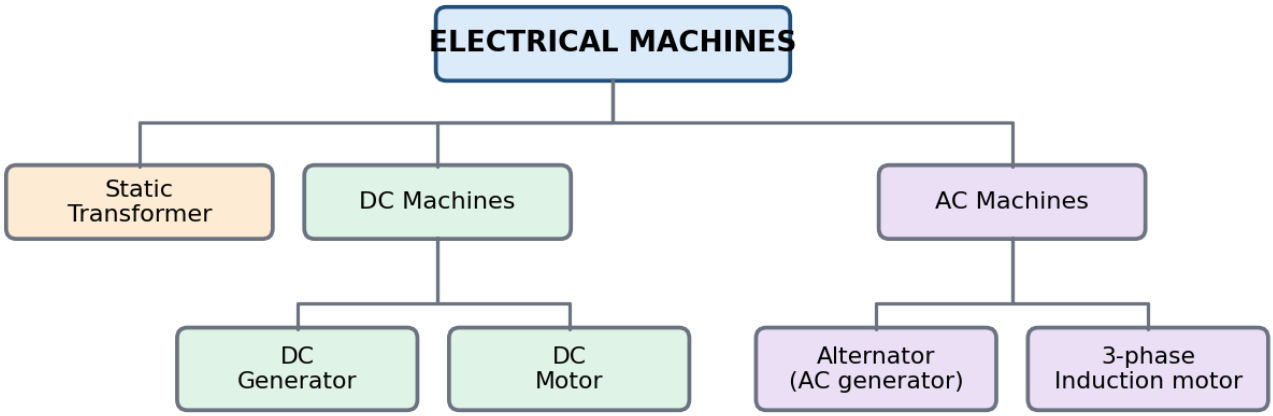


Fig 1 Classification of the machines in this unit

1 ELECTRICAL MACHINES

BASIC PRINCIPLES

Two golden principles

① **Faraday's law (generator action):** a conductor moving in a magnetic field (or a changing flux) induces an emf: e = B l v (or e = −N dΦ/dt).

② **Force on a current-carrying conductor (motor action):** F = B I l newton.

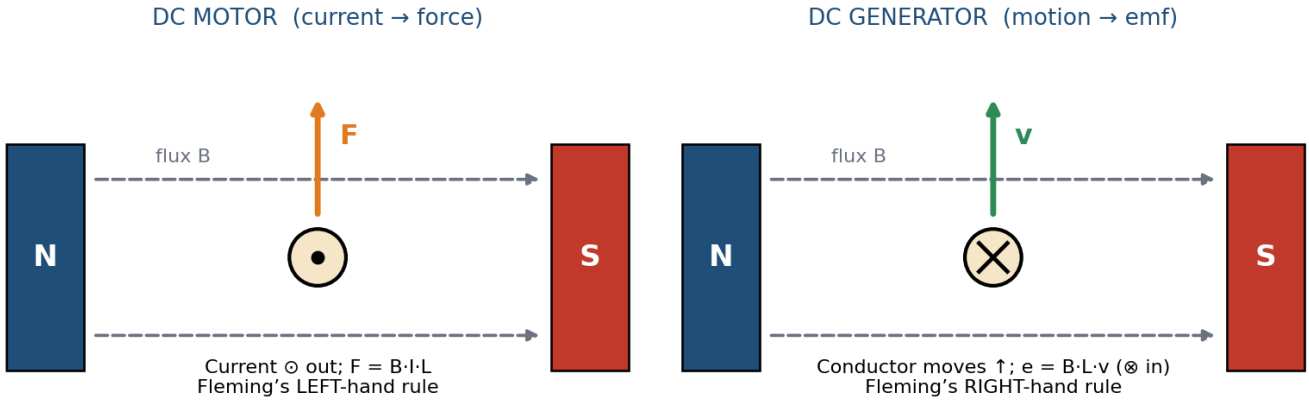


Fig 2 Motor action and generator action (same machine can do both)

1.1 DC Generator

Principle: converts mechanical energy into electrical energy (DC) by electromagnetic induction. The armature conductors cut the flux of the field poles and an emf is induced.

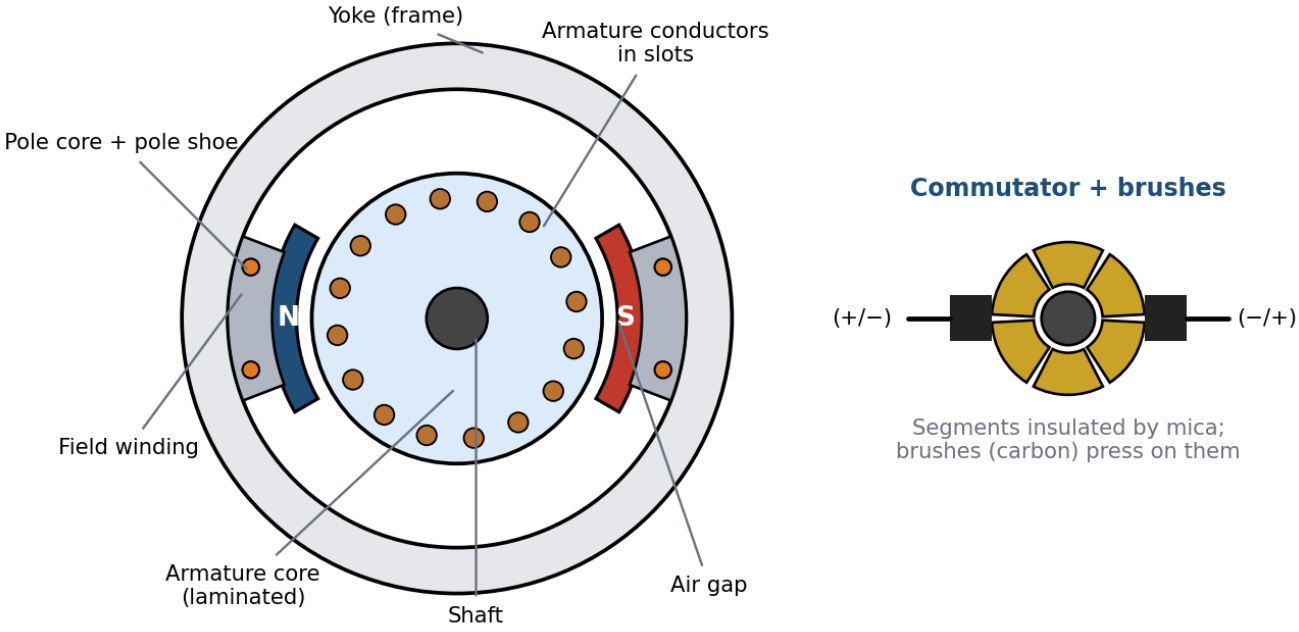


Fig 3 Cross-section of a DC machine and its commutator (same construction for generator and motor)

Part	Function
Yoke (frame)	Outer cast-iron/steel body; mechanical support and path for magnetic flux
Pole core + pole shoe	Carries field winding; shoe spreads flux evenly over the armature
Field winding	Copper coils on poles; produce the main magnetic field
Armature core + winding	Laminated steel core with slots; conductors in which emf is induced
Commutator	Copper segments; converts the alternating emf in the armature into **unidirectional** (DC) output
Brushes	Carbon blocks that collect current from the rotating commutator

$$E = (P \Phi Z N) / (60 A) \text{ volts}$$

P = number of poles, Φ = flux per pole (Wb), Z = total armature conductors, N = speed (rpm), A = number of parallel paths (A = P for lap winding, A = 2 for wave winding).

Type	Field connection	Typical applications
Separately excited	Field from an external DC source	Precise voltage control, test labs
Shunt	Field in **parallel** with armature	Battery charging, lighting, general supply
Series	Field in **series** with armature	Arc lamps, boosters
Compound	Both shunt and series fields	Lighting + power supply, welding sets

1.2 DC Motor

DEFINITION

Principle: when a current-carrying armature conductor is placed in a magnetic field it experiences a force; the force on the two sides of a coil forms a **torque** that rotates the armature. (Direction by Fleming's **left-hand rule**.)

$$T \propto \Phi I_a \quad E_b = V - I_a R_a \quad N \propto E_b / \Phi$$

Back emf (E_b): the rotating armature also generates an emf that *opposes* the supply voltage. It limits the armature current and makes the motor self-regulating — at start $E_b = 0$ so the current is very high, hence a **starter** is needed for large motors.

Motor	Characteristic	Applications
Shunt	Nearly **constant speed**	Lathes, fans, blowers, pumps, machine tools
Series	**High starting torque**; speed falls with load; never start on no-load (runaway)	Electric trains, cranes, hoists, elevators
Compound	Good starting torque + fairly steady speed	Rolling mills, presses, shears, punches

1.3 Single-Phase Transformer

DEFINITION
A **transformer** is a **static** device that transfers electrical energy from one circuit to another through **mutual induction**, changing voltage and current levels but keeping the **frequency** the same. It does **not** work on DC.

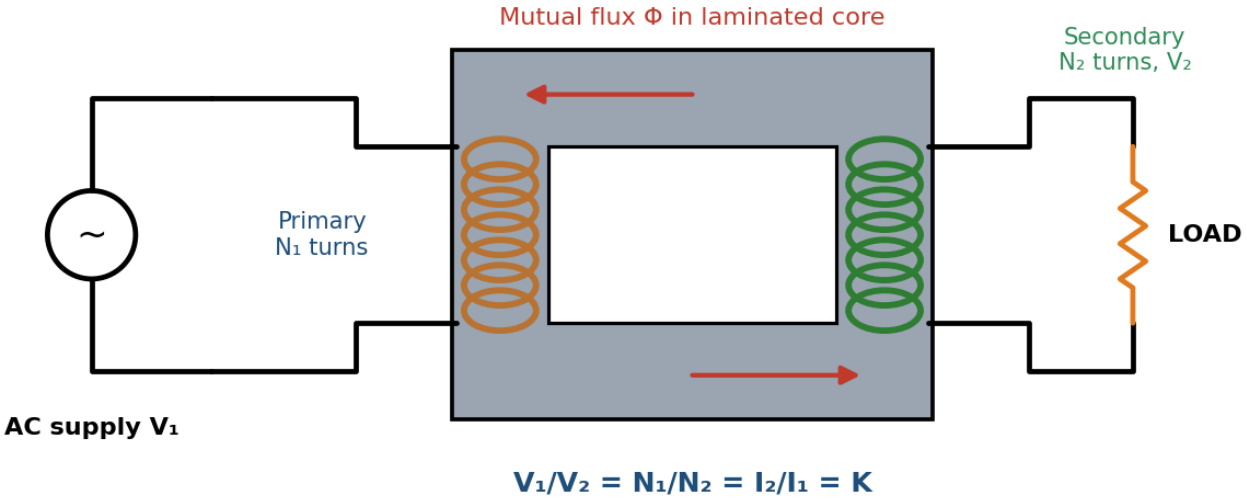


Fig 4 Single-phase transformer

Construction: a core of thin **laminated silicon-steel sheets** (reduces eddy-current loss) with two windings: **primary** (connected to supply) and **secondary** (connected to load). Types: *core type* and *shell type*. Small units are air-cooled; large ones are **oil-immersed**.
Working: AC in the primary produces an alternating flux Φ in the core → this flux links the secondary → emf is induced in the secondary by mutual induction.

$$E = 4.44\, f\, N\, \Phi_m \qquad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = K$$

Efficiency η = (Output power / Input power) × 100

Step-up transformer	Step-down transformer
$N_2 > N_1$, $K > 1$, $V_2 > V_1$, $I_2 < I_1$	$N_2 < N_1$, $K < 1$, $V_2 < V_1$, $I_2 > I_1$
Used at power stations (11 kV → 400 kV)	Used in distribution (11 kV → 415/230 V), adapters

REMEMBER
Losses: *Iron loss* (hysteresis + eddy current, constant) and *copper loss* (I^2R, varies with load). Ideal transformer: $V_1I_1 = V_2I_2$.

SOLVED EXAMPLE
Q1. A 2200/220 V transformer has 1000 primary turns. Find secondary turns and the primary current when it supplies 10 A.
$N_2 = N_1 \times V_2/V_1 = 1000 \times 220/2200 =$ **100 turns**. $I_1 = I_2 \times (V_2/V_1) = 10 \times 0.1 =$ **1 A**.

APPLICATIONS
Power transmission & distribution, mobile chargers, isolation, impedance matching, welding, voltage stabilisers.

1.4 Three-Phase Induction Motor

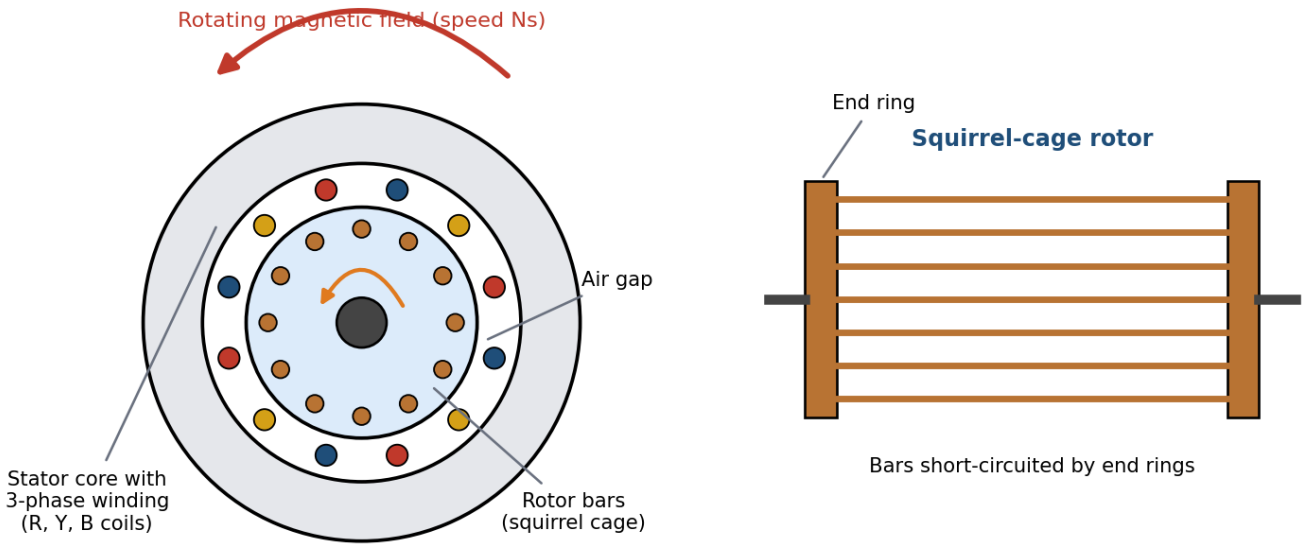


Fig 5 Stator and squirrel-cage rotor of a three-phase induction motor

DEFINITION

Principle: A three-phase supply to the stator winding creates a **rotating magnetic field** of synchronous speed N_s. It cuts the rotor conductors and induces emf → current → force (Lenz's law) → the rotor turns **in the direction of the field**, but always slightly slower. No electrical connection to the rotor, so it is also called an **asynchronous** motor.

$$N_s = 120\,f / P \text{ (rpm)} \quad \text{Slip } s = (N_s - N)/N_s$$

	Squirrel-cage rotor	Slip-ring (wound) rotor
Rotor	Copper/aluminium bars shorted by end rings	3-phase winding brought out to slip rings
Starting torque	Low	High (external resistance added)
Cost / maintenance	Cheap, rugged, no brushes	Costlier, needs brush maintenance
Uses	Pumps, fans, compressors, conveyors (≈90% of motors)	Cranes, lifts, crushers

SOLVED EXAMPLE

Q2. A 4-pole, 50 Hz induction motor runs at 1440 rpm. Find slip.

$N_s = 120 \times 50/4 = 1500$ rpm; $s = (1500 - 1440)/1500 = 0.04 =$ **4 %**.

1.5 Alternator (Synchronous Generator)

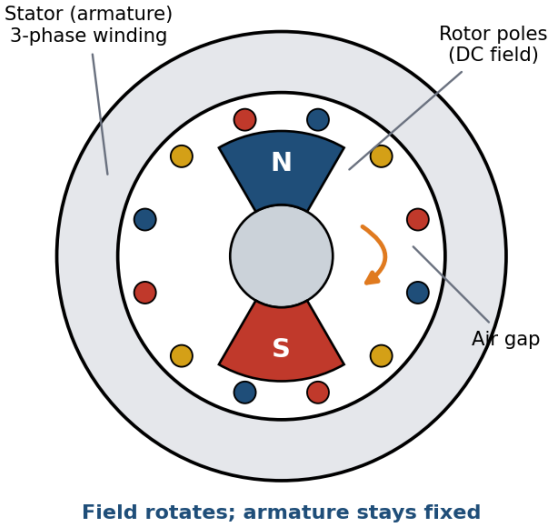


Fig 6a Cross-section of an alternator

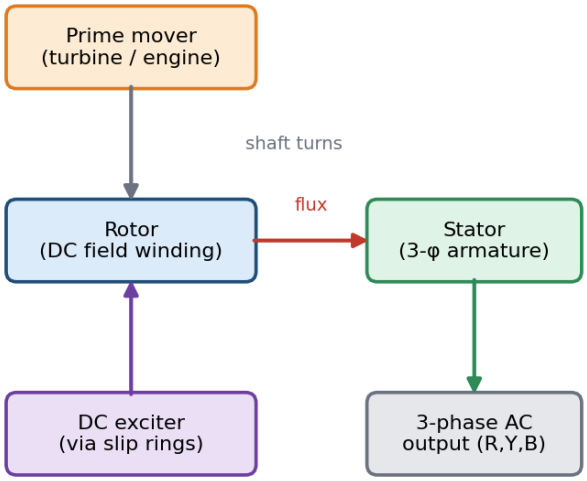


Fig 6b Block diagram of working

DEFINITION

Principle: Electromagnetic induction. The **rotor** carries a DC-excited field winding (supplied through slip rings); when turned by a prime mover its flux cuts the **stationary armature (stator)** conductors and induces a 3-phase AC emf.

$$f = P\,N / 120\ (Hz)\quad E_{ph} = 4.44\,K\,f\,\Phi\,T_{ph}$$

- **Why stationary armature?** Heavy high-voltage output (11 kV+) is easier to insulate and tap without slip rings; slip rings carry only the small DC field current.
- **Rotor types:** *Salient-pole* (low speed, hydro plants, many poles) and *cylindrical/non-salient* (high speed, steam/gas turbines, 2–4 poles).
- **Synchronous speed:** the rotor must run at N = 120f/P to deliver the correct frequency (50 Hz).

1.6 APPLICATIONS OF ELECTRICAL MACHINES

Machine	Typical applications
DC generator	Battery charging, electroplating, excitation of alternators, welding
DC motor	Electric traction (trains, trams, EVs), cranes, elevators, rolling mills, printing presses
Transformer	Power grid, adapters, inverters, isolation, instrument transformers
3-phase induction motor	Industrial drives, water pumps, fans, compressors, lifts, conveyors, washing machines
Alternator	Thermal, hydro and nuclear power stations, diesel generator sets, wind turbines, vehicle charging (alternator)

2 MEASURING INSTRUMENTS

Instruments are classified as **indicating** (show value on a scale), **recording** and **integrating** (energy meters). Every deflection instrument needs three torques: **deflecting**, **controlling** (spring) and **damping** (to stop oscillation).

2.1 Permanent Magnet Moving Coil (PMMC) Instrument

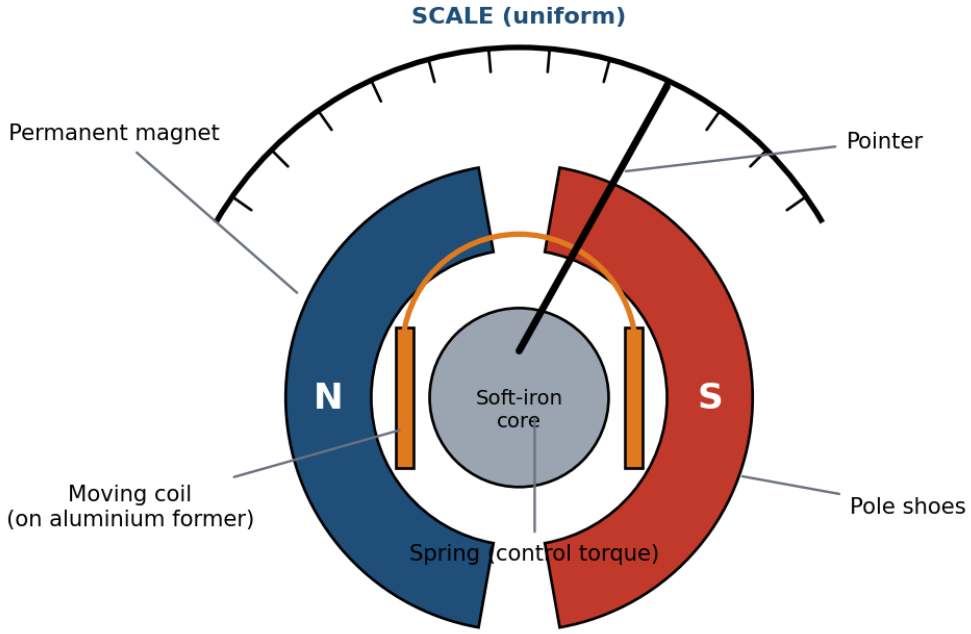


Fig 7 PMMC instrument

DEFINITION
Principle: a current-carrying coil placed in the magnetic field of a permanent magnet experiences a torque ∝ current. Deflection θ ∝ I, so the scale is **uniform**. Controlling torque: spiral **springs**. Damping: **eddy currents** in the aluminium former.

- **Advantages:** very accurate, uniform scale, high sensitivity, low power consumption, not affected by stray fields.
- **Disadvantages:** works on **DC only** (on AC the torque reverses and the pointer just vibrates), costlier, delicate.
- **Use:** DC ammeter (with parallel **shunt**) and DC voltmeter (with series **multiplier** resistor).

2.2 Moving Iron (MI) Instrument

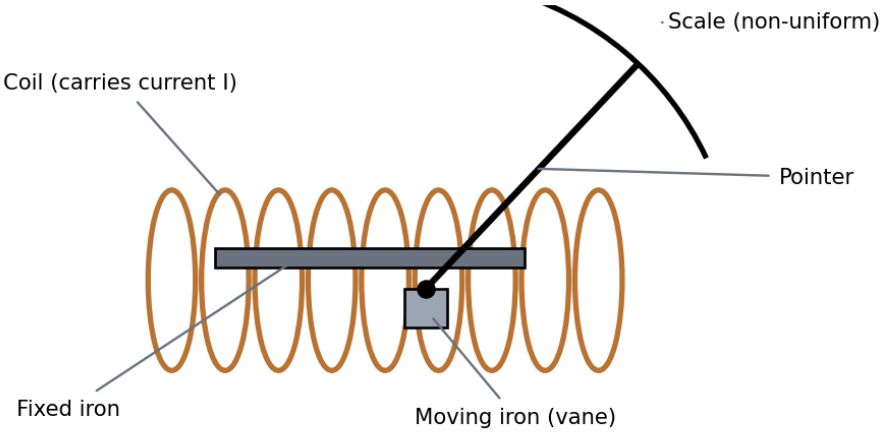


Fig 8 Moving-iron (attraction type) instrument

DEFINITION
Principle: a soft-iron piece placed in the field of a current-carrying coil gets magnetised and is **attracted** (attraction type) or two similar irons **repel** (repulsion type). Deflecting torque $\propto I^2$, so the scale is **non-uniform** (crowded at the start) but the instrument works on **both AC and DC**. Damping: air friction.

Feature	PMMC	Moving iron
Works on	DC only	AC and DC
Deflection $\propto$	I (uniform scale)	I^2 (non-uniform scale)
Accuracy / cost	High / costly	Moderate / cheap, rugged
Power consumption	Very low	Higher
Damping	Eddy current	Air friction

2.3 Wheatstone Bridge

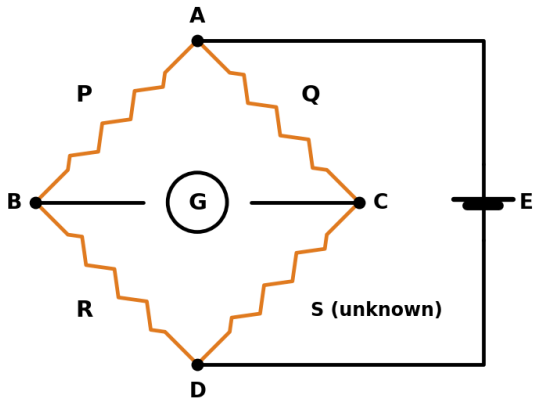


Fig 9 Wheatstone bridge

Use: measuring an unknown medium resistance accurately by comparison with known resistances.

Balance: adjust R until the galvanometer shows **zero** deflection. Then $V_B = V_C$ and the ratio arms satisfy the condition below.

P, Q = ratio arms; R = variable known resistance; S = unknown resistance.

$$P / Q = R / S \Rightarrow S = R \times Q / P$$

SOLVED EXAMPLE
Q3. At balance P = 100 Ω, Q = 1000 Ω, R = 35 Ω. Then S = 35 × 1000/100 = **350 Ω**.

- **Applications:** precise resistance measurement, strain gauges, temperature (RTD) and sensor circuits.
- **Limitations:** unsuitable for very low or very high resistance; heating by current may alter resistance.

3 QUICK REVISION & EXAM CORNER

Topic	Remember
DC generator emf	E = PΦZN/60A; commutator gives DC
DC motor	Eb = V − IaRa; T ∝ ΦIa; series = high starting torque
Transformer	$V_1/V_2 = N_1/N_2 = I_2/I_1$; E = 4.44fNΦm; no DC, no frequency change
Induction motor	Ns = 120f/P; slip = (Ns − N)/Ns; rotor never reaches Ns
Alternator	f = PN/120; rotating field, stationary armature
PMMC vs MI	PMMC: DC, uniform scale; MI: AC/DC, I^2 scale
Wheatstone	P/Q = R/S at null deflection

Short questions (2 marks)

- State the working principle of a DC motor and name Fleming's rule used.
- What is the function of a commutator?
- What is back emf? Why is a starter needed for a DC motor?
- Why is the transformer core laminated?
- Define slip of an induction motor.
- Why do alternators have a stationary armature?
- Give two advantages of PMMC instruments.
- Write the balance condition of a Wheatstone bridge.

Long questions (5–10 marks)

- Explain construction and working of a DC generator with a neat diagram; derive its emf equation.
- Explain the construction and working principle of a single-phase transformer; derive E = 4.44 f N Φm.
- Explain the production of a rotating magnetic field and working of a three-phase induction motor.
- Describe the construction and working of an alternator.
- With neat diagrams compare PMMC and moving-iron instruments.
- Explain the Wheatstone bridge and derive its balance condition.

Practice numericals (with answers)

No.	Problem	Answer
1	A 6-pole DC generator runs at 1000 rpm, Φ = 0.02 Wb, Z = 300, lap winding. Find emf.	E = 6×0.02×300×1000/(60×6) = 100 V
2	A transformer has N_1 = 500, N_2 = 50, V_1 = 440 V. Find V_2.	44 V
3	A 6-pole, 50 Hz induction motor has slip 5 %. Find rotor speed.	N_s = 1000 rpm; N = 950 rpm
4	Alternator with 4 poles must give 50 Hz. Find speed.	1500 rpm
5	DC motor: V = 230 V, I_a = 20 A, R_a = 0.5 Ω. Find E_b.	220 V

Quick MCQs

Question	Options	Ans
1. A transformer works on	(a) DC (b) AC (c) both (d) neither	b
2. Induction motor runs at … synchronous speed	(a) above (b) equal to (c) below (d) double	c
3. Speed of a series motor on no-load is	(a) zero (b) constant (c) dangerously high (d) low	c
4. Scale of an MI instrument is	(a) uniform (b) non-uniform (c) logarithmic (d) none	b

UNIT 3

Energy Resources, Electricity Bill & Safety Measures

Topics: Conventional & non-conventional resources • Hydel, nuclear, solar & wind generation • Appliance power ratings • Unit (kWh) • Two-part tariff • Bill calculation • Fuse & MCB • Electric shock • Earthing • Safety precautions

START HERE
Big picture: Electricity must be **generated** from some energy resource, **paid for** by consumers (the electricity bill) and **used safely**. This unit covers all three.

1 ENERGY RESOURCES

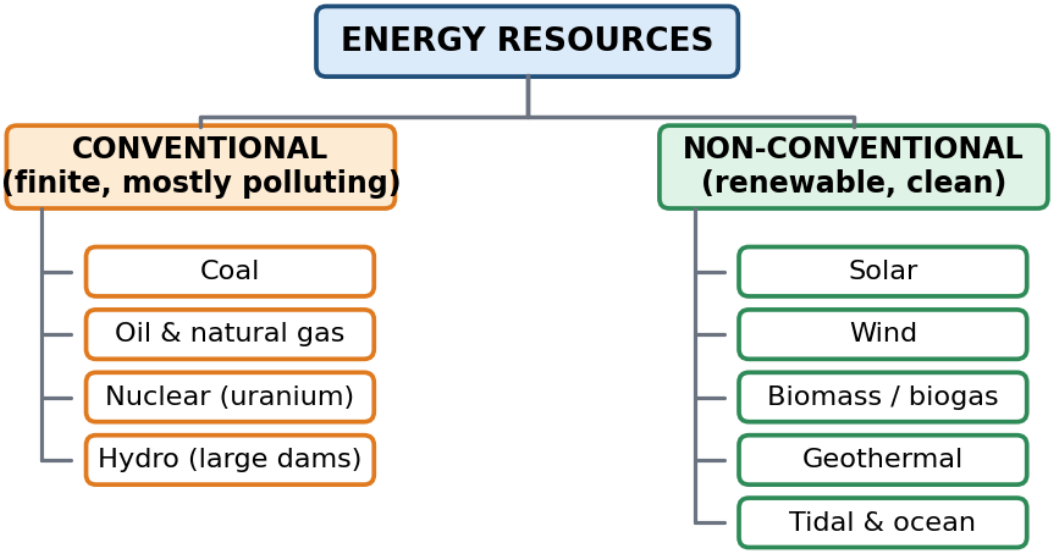


Fig 1 Classification of energy resources

Feature	Conventional	Non-conventional
Availability	Limited (fossil fuels, uranium); being depleted	Unlimited; renewable
Pollution	High (CO_2, SO_2, ash, nuclear waste)	Very low / nil
Cost	High running cost, low set-up cost (thermal)	High set-up cost, almost free fuel
Power level	Large, steady, reliable	Smaller, depends on weather/location
Examples	Coal, oil, gas, nuclear, large hydro	Solar, wind, biomass, geothermal, tidal

1.1 Hydel (Hydroelectric) Power Generation

DEFINITION

Working idea: potential energy of water stored at a height → kinetic energy as it falls through the **penstock** → mechanical energy in the **turbine** → electrical energy in the **generator**.

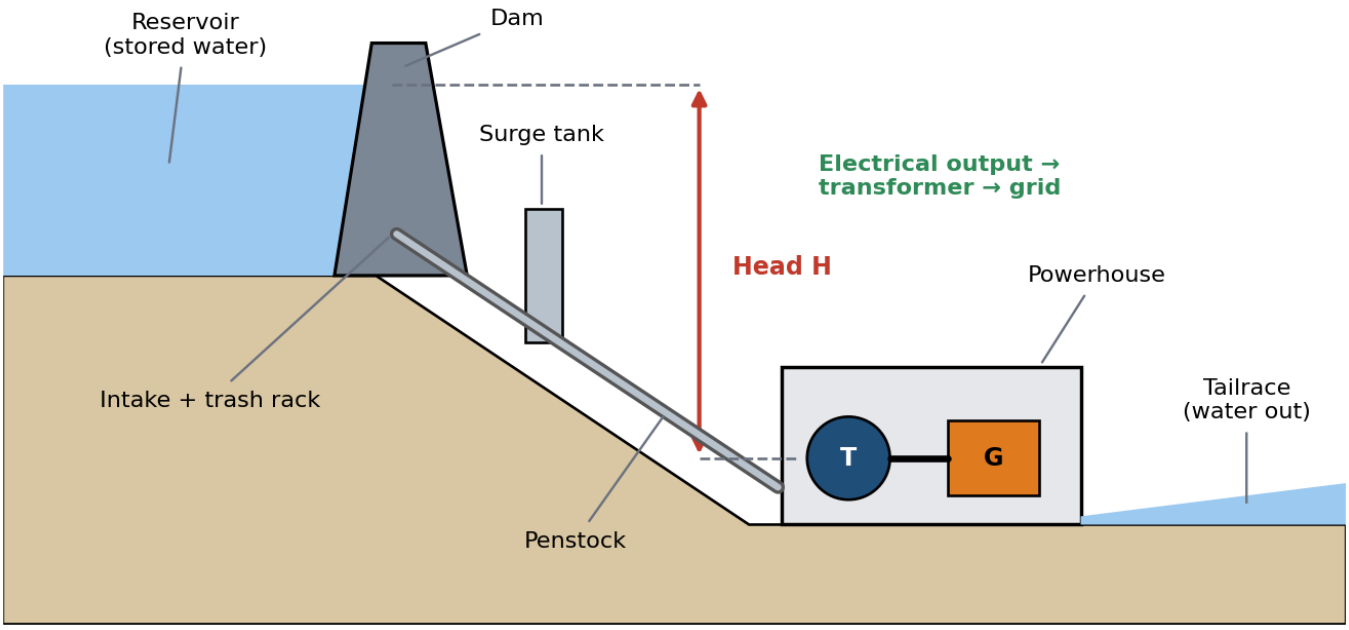


Fig 2 Layout of a hydroelectric power plant

$P = 9.81 \times Q \times H \times \eta$ kW
(Q = flow in m³/s, H = head in m, η = overall efficiency)

Component	Function
Dam + reservoir	Stores water and creates the head H
Intake / trash rack	Admits water and filters out debris
Penstock	Steel pipe carrying water to the turbine
Surge tank	Absorbs pressure surges (water hammer) when load changes
Turbine	**Pelton** (high head), **Francis** (medium head), **Kaplan** (low head)
Tailrace	Channel returning used water to the river

- **Merits:** no fuel cost, clean, long life, quick start, also gives irrigation and flood control.
- **Demerits:** very high initial cost, depends on rainfall, needs large land area, displaces people/ecology.

1.2 Nuclear Power Generation

DEFINITION

Working idea: In the reactor, neutrons split enriched **uranium-235** atoms (**nuclear fission**); each fission releases heat and more neutrons, sustaining a **chain reaction**. The heat makes steam, which drives a turbine-generator, just like a thermal plant but with the reactor replacing the boiler.

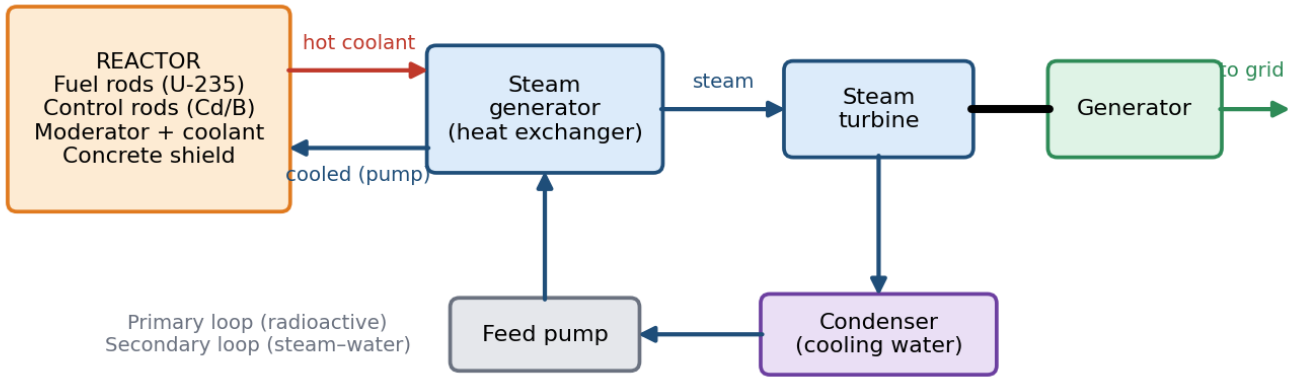


Fig 3 Block diagram of a nuclear power plant

Part	Role
Fuel rods	Enriched uranium (U-235) pellets; source of fission heat
Moderator	Slows neutrons so they cause fission (heavy water, graphite, ordinary water)
Control rods	Cadmium / boron absorb neutrons; inserted to slow or stop the reaction
Coolant	Water / heavy water / CO_2 carries heat to the steam generator
Shielding	Thick concrete + steel to stop radiation

- **Merits:** huge energy from a tiny amount of fuel, no CO_2 during operation, steady base-load power.
- **Demerits:** radioactive waste disposal, very high capital cost, risk of accidents, strict safety needs.

1.3 Solar Power Generation

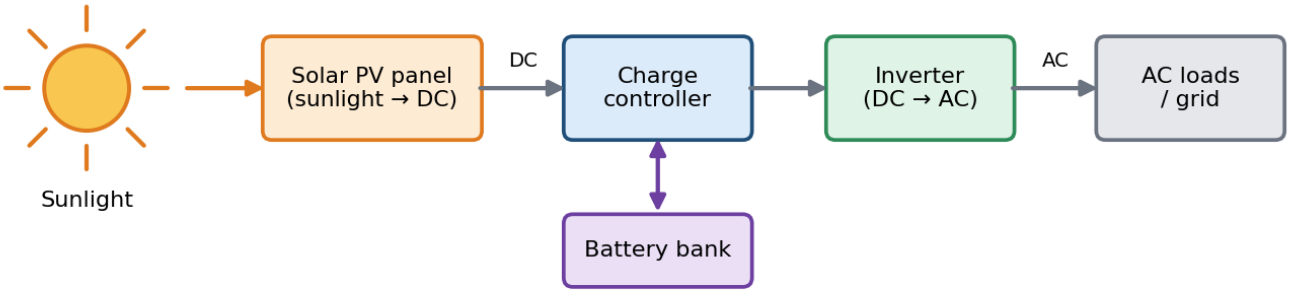


Fig 4 Stand-alone solar photovoltaic (PV) system

A **solar cell** (silicon p-n junction) converts sunlight directly into **DC** by the **photovoltaic effect**. Many cells form a module; modules form an array. The **charge controller** protects the battery, the **battery** stores energy for night use and the **inverter** converts DC to AC. In *grid-connected* systems the battery is omitted and extra power is fed to the grid.

- **Merits:** free, clean, silent, low maintenance, good for remote areas.

- **Demerits:** works only in daylight, high initial cost, needs large area, output varies with weather.

1.4 Wind Power Generation

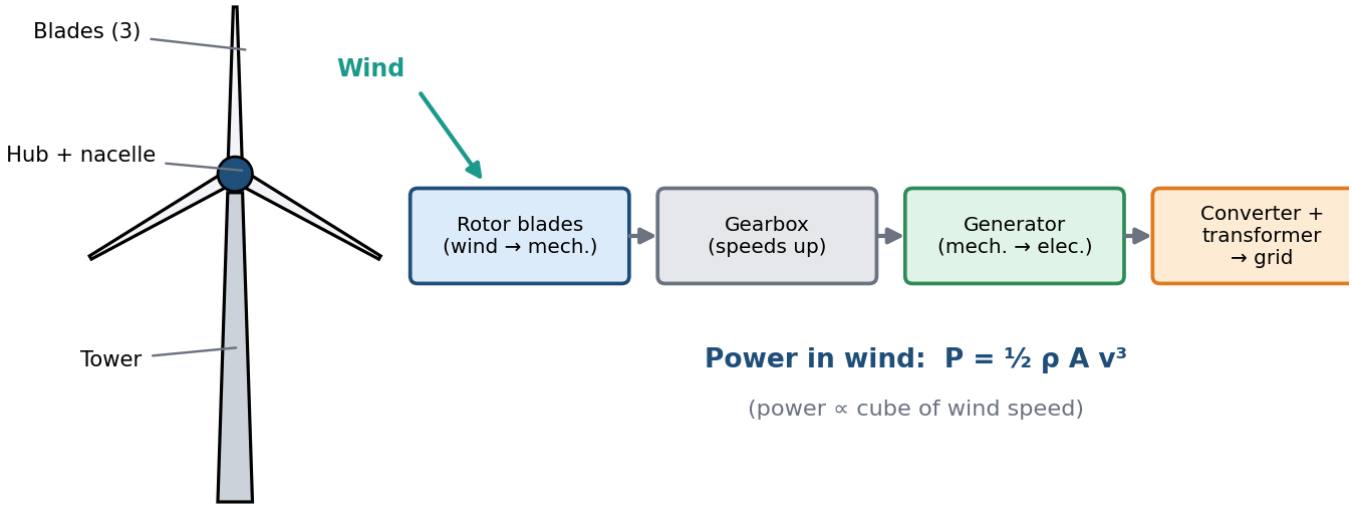


Fig 5 Wind turbine and energy-conversion chain

Moving air turns the turbine blades; the slow rotor speed is increased by a **gearbox** to drive the **generator**. A yaw system turns the nacelle to face the wind. Typical cut-in speed ≈ 3–4 m/s; best sites have steady wind > 6 m/s.

- **Merits:** free and clean fuel, land below can still be farmed, quick to install.
- **Demerits:** wind is intermittent, noise, bird hazard, needs windy sites, high initial cost.

2 ELECTRICITY BILL

2.1 Power Rating of Household Appliances

The **power rating** (in watt, W) is printed on every appliance. Typical (approximate) values:

Appliance	Typical power	Appliance	Typical power
LED bulb	5 – 12 W	Refrigerator	150 – 250 W
Tube light (LED)	20 – 40 W	Washing machine	400 – 600 W
Ceiling fan	60 – 80 W	Electric iron	750 – 1000 W
LED TV	60 – 150 W	Water heater (geyser)	1500 – 3000 W
Desktop PC (with monitor)	150 – 300 W	Microwave oven	1000 – 1400 W
Laptop	40 – 65 W	Air conditioner 1.5 ton	1200 – 1800 W
Printer (laser, printing)	300 – 500 W	Wi-Fi router	5 – 15 W

2.2 The “Unit” of Electrical Energy

DEFINITION
1 unit = 1 kilowatt-hour (kWh) = the energy used by a 1000 W (1 kW) load running for 1 hour. (1 kWh = 3.6×10^6 joule.)

Energy (kWh) = Power (W) × Time (h) / 1000



Fig 6 Steps to calculate an electricity bill

2.3 Two-Part Tariff

The consumer pays two parts: **(1) Fixed (demand) charge** – depends on connected load/sanctioned load in kW, charged per month whether or not energy is used; **(2) Energy charge** – depends on units (kWh) consumed, charged at a rate per unit (often in slabs).

Bill = Fixed charge + (Units consumed × Rate per unit) [+ taxes/duty]

SOLVED EXAMPLE
Q1. Monthly (30-day) bill for a house with: 6 LED lamps (10 W, 5 h/day), 3 fans (75 W, 8 h/day), 1 TV (100 W, 4 h/day), 1 refrigerator (200 W, runs 8 h/day), 1 laptop (60 W, 5 h/day). *Assume* fixed charge ₹100/month, energy charge ₹6/unit.
Lamps: 6×10×5/1000 = 0.30 kWh | Fans: 3×75×8/1000 = 1.80 | TV: 100×4/1000 = 0.40 | Fridge: 200×8/1000 = 1.60 | Laptop: 60×5/1000 = 0.30
Daily = **4.40 kWh** → monthly = 4.4 × 30 = **132 units** → energy charge = 132 × 6 = ₹792 → **Bill = 792 + 100 = ₹892**.
Q2. A 1.5 kW air conditioner runs 8 h/day for 30 days: 1.5 × 8 × 30 = **360 units** (≈ ₹2160 at ₹6/unit).

NOTE
Tariff rates differ from state to state and change often — use the rates given in the question or on your own bill. Saving tip: use 5-star appliances and LED lamps, switch off standby devices.

3 EQUIPMENT SAFETY MEASURES

3.1 Fuse

DEFINITION
A **fuse** is a protective device with a thin wire (element) of low-melting alloy (tin-lead, zinc, silver, copper) connected **in series**. On excessive current, the **heating effect (I^2R)** melts the element and breaks the circuit.

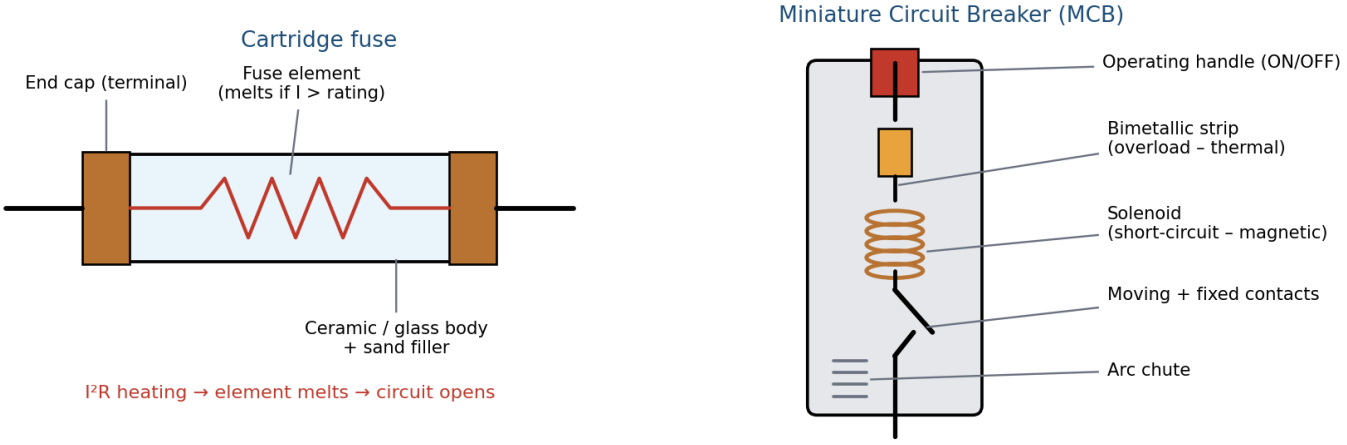


Fig 7 Cartridge fuse (left) and Miniature Circuit Breaker (right)

- **Types:** rewirable (kit-kat) fuse, cartridge (HRC) fuse.
- **Merits:** cheap, simple, no maintenance, very fast on heavy short circuits.
- **Demerits:** must be **replaced** after blowing, slow for small overloads, wrong wire rating can be dangerous, cannot be reset or remotely operated.

3.2 Miniature Circuit Breaker (MCB)

DEFINITION
An **MCB** is an automatic switch that **trips** on overload or short circuit and can be **reset** after the fault is cleared. **Overload:** heated **bimetallic strip** bends and releases the latch (delayed). **Short circuit:** the large current energises a **solenoid** whose plunger trips the contacts instantly. The arc formed is quenched in the **arc chute**.

Type	Trips at (× rated current)	Used for
B	3 – 5 ×	Lighting, resistive loads (homes)
C	5 – 10 ×	Small motors, ACs, mixed loads
D	10 – 20 ×	Transformers, large motors

	Fuse	MCB
Operation	Melts – one-time	Trips – resettable
Response	Slower on small overload	Fast, thermal + magnetic
Cost	Low	Higher
Fault indication	Need to inspect element	Handle position shows
Re-use	Replace element	Switch ON again

3.3 Personal Safety: Electric Shock

DEFINITION

Electric shock occurs when a part of the body completes a path between a live conductor and earth (or between two conductors) so that current flows through it. The **current** (not voltage alone) decides the damage: $I = V / R_{body}$.

Current through body (approx.)	Effect
≈ 1 mA	Threshold of feeling (tingling)
≈ 10 mA	Painful, muscles contract – may be unable to let go
≈ 30 mA	Breathing difficulty
50 – 100 mA	Ventricular fibrillation – may be fatal
> 200 mA	Severe burns, heart stops

Severity depends on: magnitude and duration of current, path through the body (hand-to-hand through chest is worst), skin condition (wet skin resistance ≈ 1 kΩ vs dry ≈ 100 kΩ) and supply type.

3.4 Earthing

DEFINITION

Earthing is the connection of the **non-current-carrying metal parts** of equipment (body, frame) to the general mass of earth through a low-resistance conductor. If insulation fails, the fault current flows safely to earth and the fuse/MCB trips — the body never stays at dangerous voltage.

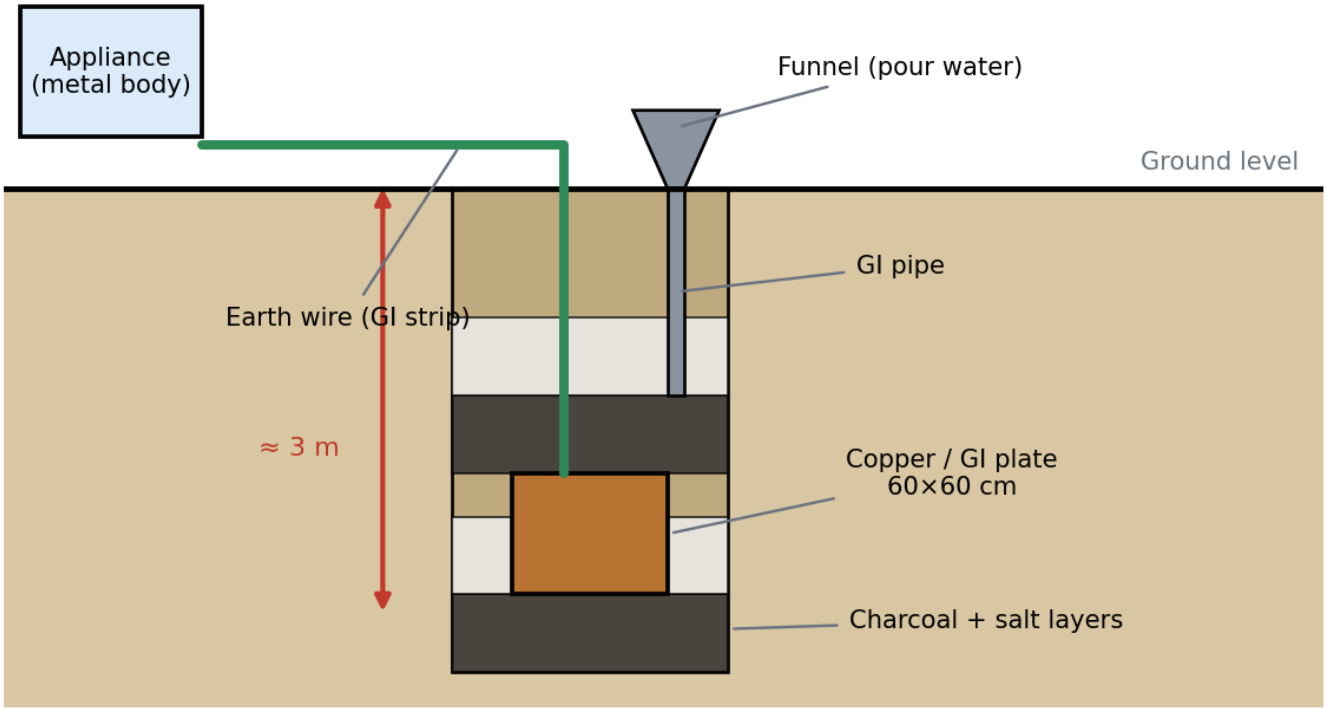


Fig 8 Plate earthing

Type	Details	Typical use
Plate earthing	Copper (60×60×3 mm) or GI (60×60×6 mm) plate buried ≈ 3 m deep with alternate layers of charcoal and salt	Substations, large buildings
Pipe earthing	GI pipe (≈ 38 mm dia, ≈ 2.5 m long) driven vertically; water poured through funnel to keep soil moist	Houses, small installations

Type	Details	Typical use
Rod earthing	Copper/GI rod (12–16 mm dia) driven into ground	Residential, telecom
Strip / wire earthing	Strip buried in trench ≈ 0.5 m deep	Rocky soil, overhead lines

REMEMBER
Earth resistance should be very low (typically < 5 Ω for homes). **Wire colours (India):** Live – Red/Brown, Neutral – Black/Blue, Earth – Green (or Green-Yellow). The earth pin of a 3-pin plug is longer and thicker.

3.5 Safety Precautions to Avoid Shock

- Always **switch off** the main supply before repairing; use a tester to confirm the line is dead.
- Use **proper earthing** and fit **MCB / ELCB (RCCB)** in every installation.
- Never touch switches or appliances with **wet hands** or while standing on a wet floor; use rubber-soled shoes / insulating mats.
- Use **ISI-marked** wires, plugs and appliances; replace damaged or bare wires immediately.
- Never use a fuse wire of the wrong rating; do not bridge a fuse with copper wire.
- Keep water away from sockets; use covered sockets in homes with children; do not overload sockets.
- If someone is stuck to a live wire: **switch off first** or push them away with a dry wooden stick – never touch them directly; then give first aid and call for medical help.

4 QUICK REVISION & EXAM CORNER

Topic	Remember
Hydel	P = 9.81 Q H η kW; Pelton – high head, Kaplan – low head
Nuclear	U-235 fission; moderator slows neutrons; control rods absorb neutrons
Solar / Wind	PV → DC → inverter; wind power ∝ v^3
Unit	1 kWh = 1000 W × 1 h; kWh = W × h / 1000
Bill	Fixed charge + units × rate
Fuse vs MCB	Fuse: I^2R melts, replace; MCB: bimetal + solenoid, resettable
Earthing	Low-resistance path for fault current; plate, pipe, rod, strip

Short questions (2 marks)

- Differentiate conventional and non-conventional energy sources.
- What is the function of the penstock and surge tank?
- What is the role of the moderator and control rods?
- Define 1 unit of electrical energy.
- What is a two-part tariff?
- Write any two demerits of a fuse.
- Why is earthing necessary?

Long questions (5–10 marks)

- Draw the layout of a hydroelectric power plant and explain its working.
- Explain the working of a nuclear power plant with a block diagram.
- Explain solar PV and wind power generation with neat diagrams.

- Explain the working of a fuse and MCB; compare them.
- What is earthing? Explain plate earthing with a neat sketch.
- Calculate the monthly bill for a given load list using a two-part tariff.

Practice numericals (with answers)

No.	Problem	Answer
1	A 2 kW heater runs 3 h daily. Find units used in 30 days.	2×3×30 = 180 units
2	A 40 W tube light and 75 W fan run 10 h daily. Units in 30 days?	(40+75)×10×30/1000 = 34.5 units
3	Fixed charge ₹120, rate ₹5/unit, consumption 200 units. Bill?	120 + 1000 = ₹1120
4	Hydel plant: Q = 20 m^3/s, H = 50 m, η = 80 %. Output power?	9.81×20×50×0.8 = 7848 kW ≈ 7.85 MW